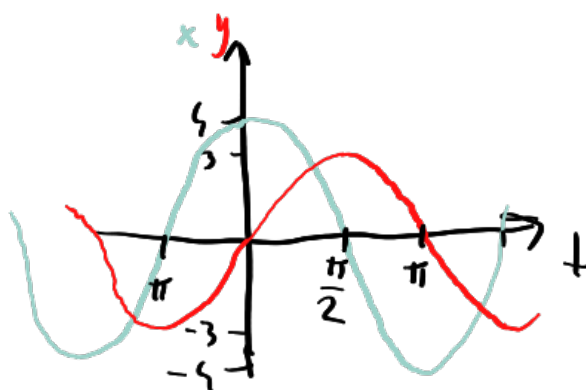


Krivulje v ravnini

- grafi
- funkcija
- parametrično $\rightarrow k = \frac{\dot{y}}{\dot{x}}$
- polarno

6. Skiciraj krivulje

a) $x(t) = 4 \cos t$
 $y(t) = 3 \sin t$
 $t \in [0, 2\pi]$

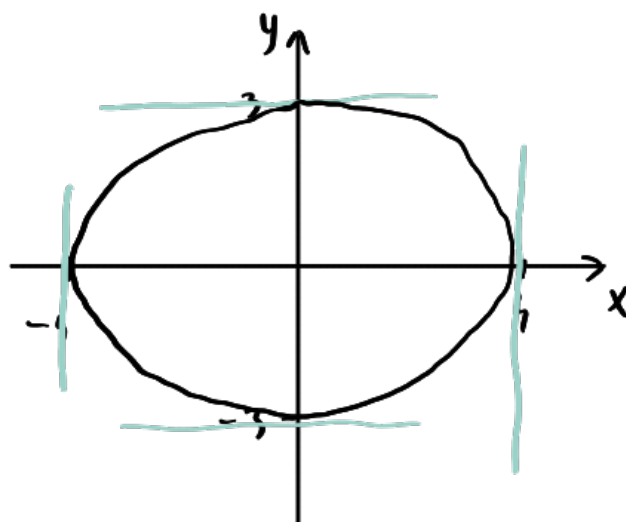


$$\dot{x} = -4 \sin t$$
$$\dot{y} = 3 \cos t$$

$$\dot{x} = 0: t = 0, \pi, 2\pi$$

$$\dot{y} = 0: t = \frac{\pi}{2}, \frac{3\pi}{2}$$

t	x	y	k
0	4	0	∞
$\frac{\pi}{2}$	0	3	0
π	-4	0	∞
$\frac{3\pi}{2}$	0	-3	0
2π	4	0	∞



$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

b) $r(\varphi) = \frac{16}{5 + 3 \cos \varphi}$

1

$$\varphi \in [0, 2\pi]$$

$$\vec{r}(\varphi) = (x(\varphi), y(\varphi)) = (r(\varphi)\cos\varphi, r(\varphi)\sin\varphi)$$

$$x(\varphi) = \frac{16\cos\varphi}{5+3\cos\varphi} \quad \dot{x} = \frac{-16\sin\varphi(5+3\cos\varphi) + 3\sin\varphi(16\cos\varphi)}{(5+3\cos\varphi)^2} = \frac{-16.5\sin\varphi}{(5+3\cos\varphi)^2}$$

$$y(\varphi) = \frac{16\sin\varphi}{5+3\cos\varphi} \quad \dot{y} = \frac{16\cos\varphi(5+3\cos\varphi) + 16\sin\varphi(3\sin\varphi)}{(5+3\cos\varphi)^2} = \frac{316 + 16.5\cos\varphi}{(5+3\cos\varphi)^2}$$

$$\dot{x} = 0: \sin\varphi = 0$$

$$\varphi = 0, \pi, 2\pi$$

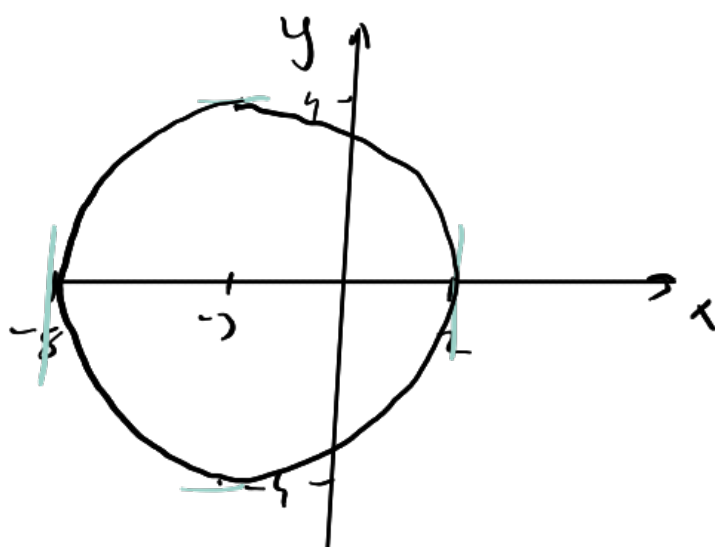
$$\dot{y} = 0: \cos\varphi = -\frac{3}{5}$$

$$\varphi_1 = \arccos(-\frac{3}{5}) + 2k\pi$$

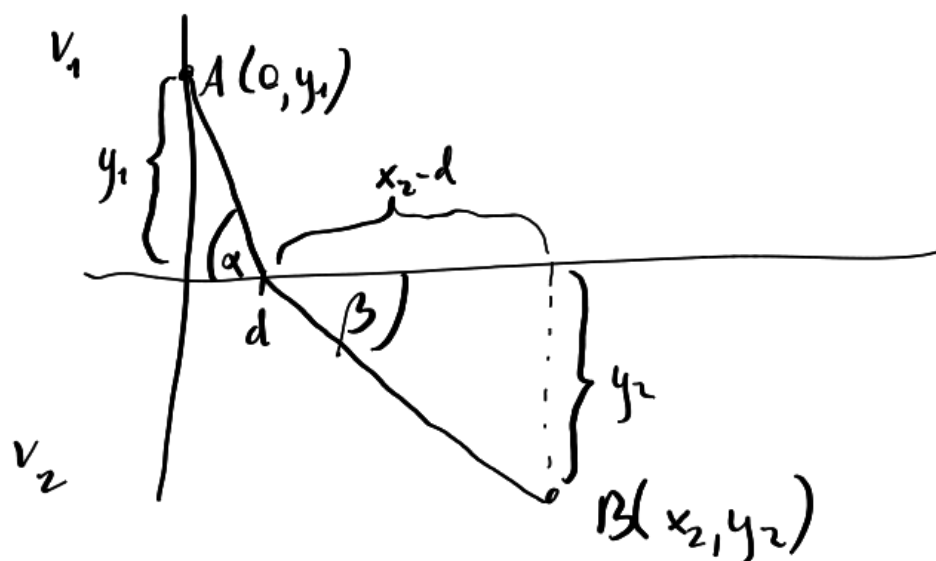
$$k \in \mathbb{Z}$$

$$\varphi_2 = -\arccos(-\frac{3}{5}) + 2k\pi$$

φ	x	y	k_{turn}
0	2	0	∞
π	-8	0	∞
2π	2	0	∞
$\arccos(-\frac{3}{5})$	-3	4	0
$2\pi - \arccos(-\frac{3}{5})$	-3	-4	0



7.



po kateri poti bo točk. telo prišlo najhitreje iz A v B?

$$t(d) = \frac{\sqrt{d^2 + y_1^2}}{v_1} + \frac{\sqrt{(x_2 - d)^2 + y_2^2}}{v_2}$$

$$t: [0, x_2] \rightarrow \mathbb{R}$$

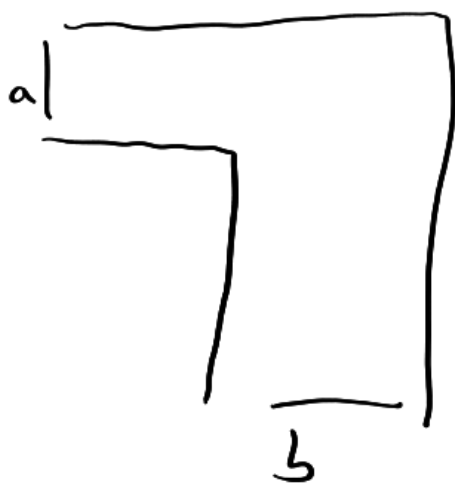
$$t'(d) = \frac{1}{v_1} (d^2 + y_1^2)^{-\frac{1}{2}} \cdot 2d + \frac{1}{v_2} ((x_2 - d)^2 + y_2^2)^{-\frac{1}{2}} \cdot (-2(d - x_2))$$

$$= \frac{d}{v_1 \sqrt{d^2 + y_1^2}} - \frac{x_2 - d}{v_2 \sqrt{(x_2 - d)^2 + y_2^2}} = 0$$

$$\frac{\sin \alpha}{v_1} = \frac{\sin \beta}{v_2}$$

$$\frac{v_1}{v_2} = \frac{\sin \alpha}{\sin \beta}$$

8.)



kot je lahko dolga lestev, da jo lahko nesemo okoli vogala

$$l(\varphi) = \frac{b}{\sin \varphi} + \frac{a}{\cos \varphi}$$

$$l'(\varphi) = 0$$

$\Downarrow$

$$l = \left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{3}{2}}$$

